

Nepal's Hydropower Gambit

Water Tower Diplomacy Between India and China

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Executive Summary

Nepal occupies one of the world's most consequential geopolitical locations: a landlocked Himalayan state of 30 million people sandwiched between the world's two most populous nations — India to the south, east, and west, and China to the north — at the fulcrum of the Indo-Pacific strategic competition. Nepal's physical geography is its defining constraint and its defining asset simultaneously: the Himalayas that isolated Nepal from external influence for centuries also endow it with one of the world's most significant untapped hydropower resources (83,000 MW of technically exploitable potential), which could transform a remittance-dependent, poverty-prone economy into a surplus electricity exporter. The paradox of Nepal's development is that it possesses extraordinary natural resource wealth (hydropower, forestry, biodiversity tourism) and extraordinary governance poverty — approximately 10 prime ministers in 10 years, a proportional representation system that fragments political power into unstable coalition governments, and endemic corruption that has prevented the infrastructure investment its hydropower potential requires.

Nepal's geopolitical position is increasingly uncomfortable: India's "special relationship" with Nepal (open border, shared currency peg history, dominant trade dependency, cultural-religious ties) is being contested by China's Belt and Road Initiative partnership with Nepal (signed 2017), which offers railway connectivity, infrastructure investment, and an alternative economic partner that reduces Nepal's total India dependency. The competition between India and China for influence in Nepal — through infrastructure investment, tourism promotion, border market development, and political relationship cultivation — is a microcosm of the broader Indo-Pacific strategic competition. Nepal, pragmatically, attempts to balance both — extracting investment commitments from each great neighbour without becoming so aligned with either that it loses access to the other.

Part I — Hydropower: The Untapped Civilisational Asset

83,000 MW and 3,000 MW Installed

Nepal's hydropower endowment is extraordinary: the combination of Himalayan rivers (Koshi, Gandaki, Karnali, Mahakali systems — fed by glaciers and monsoon runoff, dropping thousands of metres from the Himalayas to the Terai plains) provides a sustainable, renewable electricity resource estimated at 83,000 MW of technically exploitable capacity and 42,000 MW of economically viable capacity under current technology. By comparison, Nepal's current installed hydropower capacity is approximately 3,000 MW — less than 4% of its technical potential.

The Upper Tamakoshi Hydroelectric Project — Nepal's largest, at 456 MW, completed 2021 after over a decade of construction — represents what is possible when capital, engineering expertise, and political

continuity align. The project is now operational and producing power, contributing to Nepal's recent shift from a power-deficit country to a power-surplus country during high-water monsoon months. Nepal exported approximately 400 MW to India in FY2025 under a Power Purchase Agreement with India's electricity grid — the first significant cross-border power trade and a proof-of-concept for the hydropower export model.

The Hydropower Export Vision

The transformative scenario for Nepal: develop 10,000-15,000 MW of hydropower over the next 15 years, export the majority to India's electricity-hungry grid (and potentially to Bangladesh via India's transit infrastructure), and use the resulting foreign exchange earnings (\$2-4 billion annually at full development) to fund education, health, infrastructure, and economic diversification — breaking the remittance dependency that currently sustains approximately 30% of Nepal's GDP. This vision has been articulated in successive Nepalese government plans and is accepted in principle by all major political parties. The implementation challenge: large hydropower projects require billions in capital investment, years of construction, environmental impact management, and geopolitical coordination between Nepal and India (as the primary customer) that has been frustrated by repeated political transitions in Nepal and periodic India-Nepal diplomatic friction.

Part II — The India Dependency and Its Limits

Open Border, Open Vulnerability

India and Nepal share an open border — the only such border in South Asia, established by the 1950 Treaty of Peace and Friendship, which allows Indian and Nepali citizens to move freely, work, and reside in each other's countries without visas or work permits. The open border has created one of the world's most economically integrated bilateral relationships: approximately 6 million Nepalis work in India (the largest single remittance source for Nepal), Nepal imports approximately 70% of its goods from or through India, India is Nepal's largest trading partner by a large margin, and Nepali currency is informally tied to the Indian rupee at a fixed exchange rate.

India's leverage over Nepal is thus structural and pervasive: India can (and has) imposed informal blockades on goods transiting through Indian territory to Nepal, creating fuel and commodity shortages that cause severe economic disruption. The most significant blockade — September-December 2015, following the promulgation of Nepal's new constitution (which the Madhesi ethnic minority, allegedly with Indian encouragement, protested) — caused fuel shortages that shut hospitals, schools, and industry for months and produced deep Nepali resentment of Indian "big brother" behaviour. This resentment is the emotional fuel for Nepal's China engagement — not strategic calculation but a human response to coercive dependency.

The Blockade Effect on China Engagement

Nepal's acceleration of China engagement after the 2015 blockade was direct and causal. Nepal signed the Transit Transport Agreement with China in 2016 (enabling Nepal to use Chinese ports for third-country trade as an alternative to Indian sea ports), joined the BRI in 2017, and began negotiating connectivity infrastructure (road upgrades, railway extension from Kerung to Kathmandu) that would provide an alternative supply corridor to India's. As of 2026, the Chinese railway to Kathmandu remains in feasibility study and financing negotiation — the engineering challenge (crossing the world's most complex mountain geology) and the financing terms (loan rather than grant, which Nepal's debt-to-GDP sustainability framework constrains) have prevented final investment decisions.

Part III — Governance and Political Fragmentation

The Revolving Door of Prime Ministers

Nepal's political system — restructured by the 2015 constitution into a federal parliamentary republic with proportional representation for a significant share of the 275-seat House of Representatives — produces a fragmented party landscape that has made stable government nearly impossible. The main parties (CPN-UML, CPN-Maoist Centre, Nepali Congress, and smaller regional parties) are internally fissured and require multi-party coalitions to form governments. Between the 2017 elections and 2026, Nepal has had approximately 10 different prime ministers — averaging less than one year per prime minister — disrupting infrastructure project continuity, budget execution, and foreign investment negotiations.

The political instability is not merely a governance inconvenience: large hydropower projects require 5-10 year regulatory and financing certainty; foreign investors making \$500 million+ hydropower investments need confidence that the government signing the agreement will be succeeded by a government that honours it. Nepal's track record — renegotiated agreements, changed development board mandates, and cancelled licenses under successive governments — has deterred private capital that would otherwise be attracted by Nepal's hydropower economics.

Part IV — The Remittance Paradox

30% of GDP from Migration

Remittances from Nepalis working abroad contribute approximately 30% of Nepal's GDP — one of the highest ratios in the world. Approximately 4-5 million Nepalis (out of a working-age population of approximately 15 million) work abroad: in India (6 million, unreported in official statistics as the open border makes these internal migration by treaty definition), Gulf countries (Qatar, UAE, Saudi Arabia, Kuwait — approximately 2 million), Malaysia, and increasingly South Korea and Japan. The remittance inflow (approximately \$9-10 billion annually) sustains household consumption, funds children's education, builds houses, and services debt — but does not build productive capital, improve institutional quality, or diversify the economy.

The "Gulf trap": Nepal's remittance economy creates structural dependency on migrant labour that is physically dangerous (Nepali workers in Qatar construction sites for the 2022 FIFA World Cup — approximately 6,500 Nepali deaths in Qatar over the decade of construction, according to The Guardian's reporting), diplomatically complex (negotiating bilateral labour agreements with Gulf states), and economically hollow (remittances fund consumption without building domestic productive capacity). Breaking out of remittance dependency requires the domestic economic opportunities — hydropower, tourism, manufacturing — that political instability has prevented from scaling.

Conclusion: The Country That Could

Nepal's development paradox is painful precisely because the solution is visible: develop the hydropower resource, use the proceeds to fund human capital and infrastructure, and leverage the young population (median age 25) and mountaineer-grade physical endurance of Nepal's workforce into higher-value economic activity. The constraint is governance — the political system that produces revolving-door prime ministers, the corruption that siphons infrastructure investment into personal enrichment, and the geopolitical dependency (on India) and geopolitical aspiration (toward China) that consumes political attention that should be focused on domestic economic management.

Nepal is a country with a civilisational resource base — water, mountains, biodiversity, and a globally respected culture (the birthplace of Buddha, the home of Everest) — and governance institutions insufficient to realise it. The 2026 outlook: hydropower development will accelerate (the India export framework is established, the economics work), political fragmentation will persist, and Nepal will

continue its characteristic path of muddling through — performing below its potential, above its governance capacity, and maintaining the Himalayan resilience that has survived conquest, earthquake, and political chaos for centuries.

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All data from Nepal Electricity Authority, Nepal Rastra Bank, World Bank, ADB, and cited sources.